Optimizing distribution of drinking water in remote areas of Nepal

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Problem Formulation

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Goal: Distribute drinking water supply in remote areas of Nepal

Restrictions:

- All houses must have some minimum access to water
- Water amount must be proportional to their needs
- There are losses due to the terrain/pipe quality

Methods used:

- Lagrange Method
- Simulated Annealing

Lagrange Method

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a : delivery-loss factor
b : the smallest required amount of water per household
d : demand per household
w : house priority/vulnerability factor
S : total available water supply

Objective function: $\min f(x) = \sum w \cdot (d - x)^2$

Restrictions: $1 \leqslant a \leqslant 1.25,\ b > d,\ w > 0,\ S > 0$

Simulated Annealing

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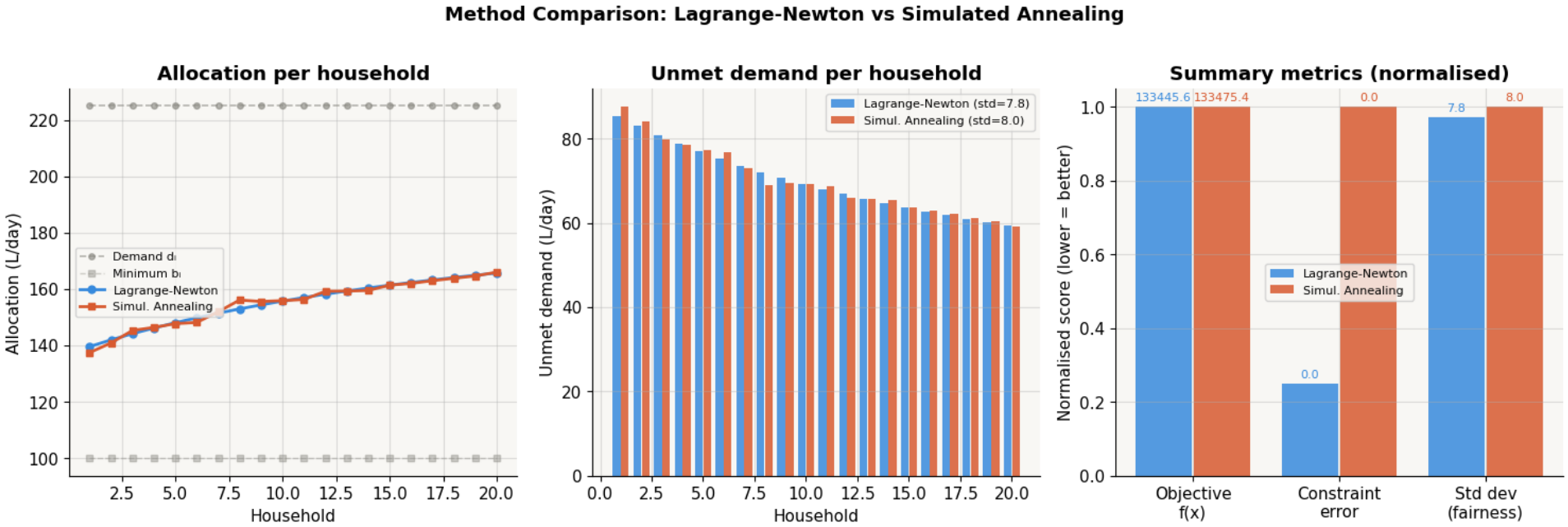
- "Mimics the physical process of annealing metal"
- Tk = T0 * (1 – k /k max) (with 50'000 Iterations or 20 L per day)
  - Different Distributions
  - Prioritizing higher vulnerability
- ai_xi = S -> remaining satisfied
- Exploration High T / low T

Method Comparison



Method Comparison

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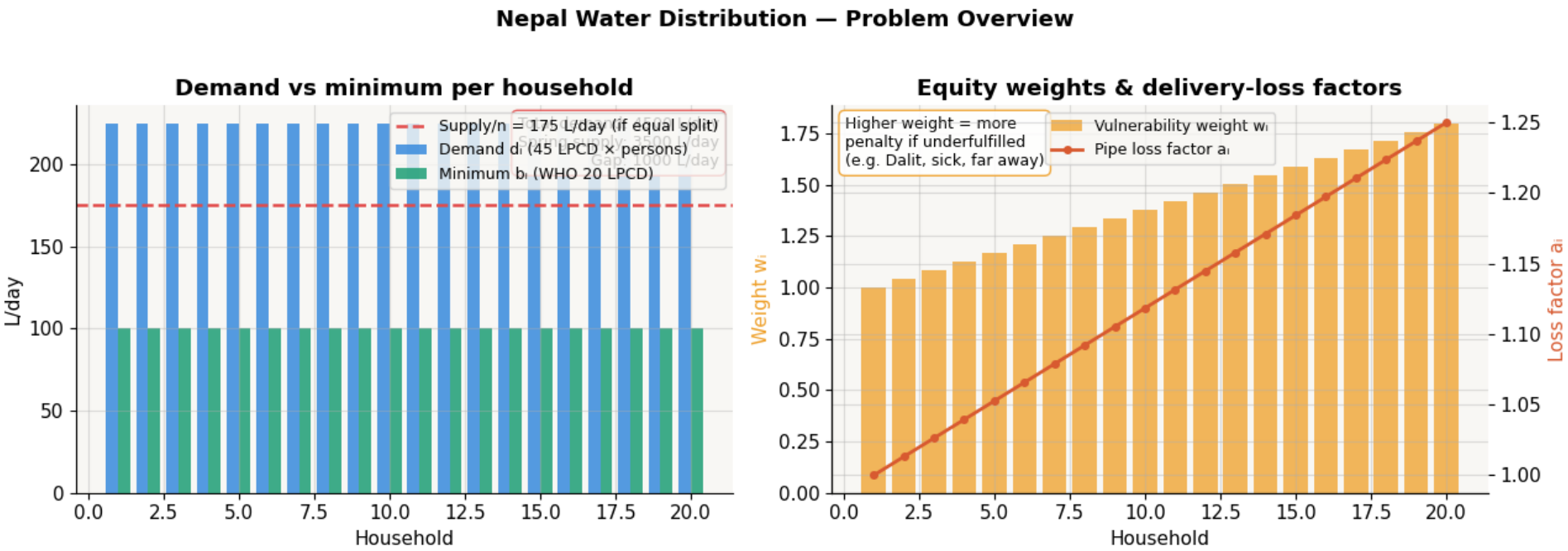

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Questions

Thank you